

08.28.24

Cognition, Communication, & Survey Measurement SURVMETH 632

Fall, 2024

Wednesday, 9:00 AM – 11:30 PM

ISR 1070, Lefrak 1208

Instructor: Aaron Maitland

Email: maitland@umd.edu

Overview of the Course

This course will cover major sources of survey measurement error from the perspective of the social sciences, particularly cognitive psychology. The first part of the course (weeks 1 – 9) focuses on reporting errors and how they arise: misunderstanding questions, forgetting relevant information, using imprecise estimation strategies, and question and response order effects; the classes in this part of the course are ordered to reflect the sequence of processes that respondents (ideally) carry out to produce an answer to a survey question. The second part of the course (Weeks 10-13) covers special topics in measurement error: Classification and Coding, Technology Mediated Responding, Survey Satisficing, and Cross-Cultural Survey Research.

Course Requirements

Grades for the course will be based on:

- Participation in class demonstrating understanding of the readings and in-class material (10% of grade)
- A mid-term exam (40% of grade)
- A final exam (50% of grade)

Note: Exams are closed book. No additional information (including online material) is allowed. Exams will be administered via Canvas, i.e., your answers are submitted through the Canvas platform. You will have 2.5 hours (without stopping) to complete the exam over a one-week window prior to the due date. You will need to sign an honor pledge to confirm that you will not use any resources beyond your knowledge.

Course materials

Required reading other than chapters from the Tourangeau et al. (2000) text will be on the Canvas course site. Some recommended, i.e., optional, readings will be posted on the site as well (preceded by “O” for optional). Lecture notes will be available on the site before each class.

Accommodations for Students with Disabilities

University of Michigan

If you believe you need an accommodation for a disability, please contact the Services for Students with Disabilities (SSD) office to help us determine appropriate academic accommodations. SSD (734-763-3000; <http://ssd.umich.edu>) typically recommends accommodations through a Verified Individualized Services and Accommodations (VISA) form. Any information you provide is private and will remain confidential.

University of Maryland

If you believe you need an accommodation for a disability, please contact the Accessibility and Disability Services (ADS) office to register for services: 301-314-7682. More information is available at <https://counseling.umd.edu/ads>.

Academic conduct

Clear definitions of the forms of academic misconduct, including cheating and plagiarism, as well as information about disciplinary sanctions for academic misconduct at the University of Michigan may be found at the Rackham web site (<https://rackham.umich.edu/academic-policies/section8/>) the University of Maryland at the web site for the Office of the President (<https://www.president.umd.edu/sites/president.umd.edu/files/documents/policies/III-100A.pdf>). Knowledge of these rules is the responsibility of the student and ignorance of them does not excuse misconduct. The student is expected to be familiar with these guidelines before submitting any written work or taking any exams in this course. Lack of familiarity with these rules in no way constitutes an excuse for acts of misconduct. Charges of plagiarism and other forms of academic misconduct will be dealt with very seriously and may result in oral or written reprimands, a lower or failing grade on the assignment, a lower or failing grade for the course, suspension, and/or, in some cases, expulsion from the university.

Attending Class

Students are expected to attend all classes in person. If you are unable to attend a particular class in person due to extenuating circumstances, you may be able to attend remotely if you communicate about your circumstances with the instructor and the instructor approves – but you must take the initiative. (Some JPSM students may be approved to attend all classes remotely.) If you have been approved to attend a class remotely, please connect using Meeting ID 914 6732 0952 and passcode 2024632. Please be sure to test Zoom on whatever computing device you plan to use. Please make sure your camera is turned on and that, when you are not speaking, your microphone is muted. You are free to use a virtual background but, again, please keep your camera turned on throughout the class. You will need a good Internet connection to participate and may need a headset to assure you can hear and be heard with adequate clarity. If you need assistance connecting to or testing Zoom, please contact Elisabeth Schneider

(dodo@umich.edu) or Patsy Gregory (pagregor@umich.edu).

Class Video Recordings

Video recordings of each class will be available shortly after class, barring technical problems. You can access them via the Zoom option in the left-side navigation bar in Canvas and then through “Cloud Computing.”

Schedule and Reading Assignments

August 28: Cognitive views of error and implications for survey response

Tourangeau, R., Rips, L. & Rasinski, K. (2000)². Chapter 1, “An Introduction and a Point of View,” *The psychology of survey response*. Cambridge: Cambridge University Press.

Recommended/Optional reading

Sudman, S., Bradburn, N. & Schwarz, N. (1996)¹. Chapter 1, “Introduction,” and Chapter 3, “Answering a Survey Question: Cognitive and Communicative Processes.” *Thinking about answers: The Application of cognitive processes to survey methodology*. San Francisco: Jossey-Bass Publishers.

For background or a refresher in cognitive psychology, see chapters 1, 6, 7 and 12 (and others) in:

Anderson, J.R. (2015). *Cognitive Psychology and its Implications, Eighth Edition*. Worth Publishers.

For background or a refresher in social cognition, see chapters 1, 2, 3, 7, 9 and 11 in:

Kunda, Z. (1999). *Social Cognition: Making Sense of People*. Cambridge, MA: MIT Press.

¹Henceforth referred to as “SB&S (1996).”

²Henceforth referred to as “TR&R (2000).”

September 4: Understanding Survey Questions: Basic Processes

Clark, H. H. & Schober, M. F. (1992). Asking questions and influencing answers. In Tanur, J. M. (Ed.), *Questions about questions: Inquiries into the cognitive bases of surveys*. New York: Russell Sage Foundation, (15-48).

Conrad, F.G., Schober, M.F., & Schwarz, N. (2014). Pragmatic processes in survey interviewing. In T. Holtgraves (Ed.), *Oxford Handbook of Language and Social Psychology* (pp. 420-437). Oxford: Oxford University of Press.

TR&R (2000). Chapter 2, "Respondents' understanding of survey questions."

Recommended Reading

Beatty, P. (1995). Understanding the standardized/non-standardized interviewing controversy. *Journal of Official Statistics*, 11, 147-160.

Conrad, F.G. & Schober, M.F. (2021). Clarifying question meaning in standardized interviews can improve data quality even though wording may change: a review of the evidence. *International Journal of Social Research Methodology*, 24:2, 203-226, DOI: [10.1080/13645579.2020.1824627](https://doi.org/10.1080/13645579.2020.1824627)

Conrad, F.G. & Schober, M.F. (2000). Clarifying question meaning in a household telephone survey. *Public Opinion Quarterly*, 64, 1-28.

Fowler Jr, F. J., & Mangione, T. W. (1990). *Standardized survey interviewing: Minimizing interviewer-related error* (Vol. 18). Sage.

Schober, M.F., Conrad, F.G. and Fricker, S.S. (2004). Misunderstanding standardized language in research interviews. *Applied Cognitive Psychology*, 18, 169-188.

Suchman, L., & Jordan, B. (1990). Interactional troubles in face-to-face survey interviews. *Journal of the American Statistical Association*, 85, 232-241.

Schwarz, N. (1996). Chapter 2, "Cognition and communication: The logic of conversation," and Chapter 6, "Making one's contribution informative: The changing meaning of repeated questions." *Cognition and communication: Judgmental biases, research methods, and the logic of conversation*. Mahwah, NJ: Lawrence Erlbaum Associates, Publishers.

Schwarz, N., Strack, F., & H. Mai. (1991). Assimilation and contrast effects in part-whole question sequences: A conversational logic analysis. *Public Opinion Quarterly*, 55, 3-23.

September 11: Understanding Questions in Survey Interviews

Fowler Jr., F.J., & Cannell, C.F. (1996). Using behavioral coding to identify cognitive problems with survey questions." In N. Schwarz & S. Sudman (Eds.), *Answering Questions: Methodology for Determining Cognitive and Communicative Processes in Survey Research* (pp. 15-36). San Francisco, CA: Jossey-Bass.

Schober, M.F., Conrad, F.G, Dijkstra, W., & Ongena, Y. (2012). Disfluencies and gaze aversion in unreliable responses to survey questions. *Journal of Official Statistics*, 28, 555-582.

Schaeffer, N.C & Maynard, D.W. (2008). The contemporary standardized survey interview for social research. In Conrad, F.G. and Schober, M.F. (Eds) *Envisioning the Survey Interview of the Future* (pp. 31-57). Hoboken, NJ: John Wiley & Sons.

Recommended Reading

Brennan, S. E., & Williams, M. (1995). The feeling of another's knowing: Prosody and filled pauses as cues to listeners about the metacognitive states of speakers. *Journal of Memory and Language*, 34, 383-398.

Ongena, Y. & Dijkstra, W. (2007). A model of cognitive processes and conversational principles in survey interview interaction. *Applied Cognitive Psychology*, 21, 145-163.

Schober, M.F., & Bloom, J.E. (2004). Discourse cues that respondents have misunderstood survey questions. *Discourse Processes*, 38, 287-308.

Schober, M.F. & Conrad, F.G. (1997). Does conversational interviewing reduce survey measurement error? *Public Opinion Quarterly*, 61, 576-602.

September 18: Memory, Forgetting and Measurement Error

Fisher, R. & Quigley, K. (1992). Applying cognitive theory in public health investigations: Enhancing food recall with the cognitive interview. In J. M. Tanur (ed), *Questions about questions: Inquiries into the cognitive bases of surveys*. New York: Sage (154-169).

Lee, L., Brittingham, A., Tourangeau, R., Willis, G., Ching, P., Jobe, J. & Black, S. (1999). Are reporting errors due to encoding limitations or retrieval failure? Surveys of child vaccination as a case study. *Applied Cognitive Psychology*, 13, 43-63.

Thomsen, D. K. (2015). Autobiographical Periods: A Review and Central Components of a Theory. *Review of General Psychology*, 19.

TR&R (2000). Chapter 3, "The role of memory in survey responding."

Recommended Reading

- Barsalou, L. (1988). The content and organization of autobiographical memories. In Neisser, U. and Winograd, E. (Eds.), *Remembering reconsidered: Ecological and traditional approaches to memory* (193-243).
- Belli, R. F (1998). The structure of autobiographical memory and the event history calendar: Potential improvements in the quality of retrospective reports in surveys. *Memory*, 6, 383-406.
- Belli, R.F., Shay, W. L. & Stafford, Frank, P. (2001). Event history calendars and question list surveys: A direct comparison of interviewing methods. *Public Opinion Quarterly*, 65, 45 - 74.
- Brown, N. (1998). Event cueing, event clusters, and the temporal distribution of autobiographical memories. *Applied Cognitive Psychology*, 12, 305-319.
- Conway, M.A. (2005) Memory and the self. *Journal of Memory and Language* 53, 594–628
- Loftus, E., Smith, K., Klinger, M. & Fiedler, J. (1992). Memory and mismemory for health events. In Tanur, J. M. (Ed.) *Questions about questions: Inquiries into the cognitive bases of surveys*. New York: Russell Sage Foundation (102-137).
- Rubin, D.C. (2012). The basic systems model of autobiographical memory. In Berntsen, D. & Rubin, D.C. (Eds.) *Understanding Autobiographical Memory: Theories and Approaches*. Cambridge: Cambridge University Press.
- Wagenaar, W. (1986). My memory: A study of autobiographical memory over six years. *Cognitive Psychology*, 18, 225-252.

September 25: Placing Events in Time

- Rubin, D.C. & Baddeley, A.D. (1989). Telescoping is not time compression: A model of the dating of autobiographical events. *Memory and Cognition*, 17, 653-661.
- Rips, L. J., Conrad, F.G. & Fricker, S. S. (2003). Straightening the seam effect in panel surveys. *Public Opinion Quarterly*, 67, 522-554
- TR&R (2000). Chapter 4, "Answering questions about dates and durations."

Recommended Reading

- Sudman, S., Bradburn, N. & Schwarz, N. (1996)³. Chapter 8, "Event Dating." *Thinking about answers: The Application of cognitive processes to survey methodology*. San Francisco: Jossey-Bass Publishers.
- Brown, N., Rips, L. & Shevell, S. K. (1985). The subjective dates of natural events in very long term memory. *Cognitive Psychology*, 17, 139-177.

- Freedman, V.A., Conrad, F., Cornman, J., Schwarz, N., Stafford, F. (2014). Does Time Fly When You Are Having Fun? A Day Reconstruction Method Analysis. *Journal of Happiness Studies* 15, 639–655.
- Friedman, W. J. (1993). Memory for the time of past events. *Psychological Bulletin*, 35, 44-66
- Huttenlocher, J., Hedges, L.V., & Bradburn, N. M. (1990). Reports of elapsed time: Bounding and rounding processes in estimation. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 16, 196-213.
- Lee, P. J. & Brown, N. R. (2004). The role of guessing and boundaries on date estimation biases. *Psychonomic Bulletin & Review*, 11, 748-754.

October 2: Reporting Quantities

Midterm exam available after class

- Belli, R., Schwarz, N., Singer, E. & Talarico, J. (2000). Decomposition can harm the accuracy of behavioral reports. *Applied Cognitive Psychology*, 14, 295-308.
- Conrad, F. G., Brown, N. R. & Cashman, E. R. (1998). Strategies for estimating behavioural frequency in survey interviews. *Memory*, 6, 339-366.
- TR&R (2000). Chapter 5, “Factual judgments and numerical estimates.”

Recommended Reading

- Alba, J.W., Chromiak, W., Hasher, L. & Attig, M. S. (1980). Automatic encoding of category size. *Journal of Experimental Psychology*, 6, 370-378.
- Blair, E. & Burton, S. (1987). Cognitive processes used by survey respondents to answer behavioral frequency questions. *Journal of Consumer Research*, 14, 280-288.
- Brown, N. R. (1995) Estimation strategies and the judgment of event frequency. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 21, 1539-1553.
- Brown, N. R., Williams, R.L., Barker, E.T. & Galambos, N.L. (2007) Estimating frequencies of emotions and actions: A web-based diary study. *Applied Cognitive Psychology*, 21, 259- 276.
- Brown, N., & Siegler, R. (1993). Metrics and mappings: A framework for understanding real- world quantitative estimation. *Psychological Review*, 100, 311-325.
- Burton, S. & Blair, E. (1991). Task conditions, response formulation processes, and response accuracy for behavioral frequency questions in surveys. *Public Opinion Quarterly*, 55, 50- 79.

- Conrad, F.G., Brown, N.R. & Dashen, M. (2003). Estimating the frequency of events from unnatural categories. *Memory & Cognition*, 31, 552-562
- Menon, G. & Yorkston, E. (2000). The use of memory and contextual cues in the formation of behavioral frequency judgments. In Stone, A. A., Turkkan. J.S., Bachrach, C. A., Jobe, J. B., Kurtzman, H. S. & Cain, V. S. (Eds.) *The science of self-report*. Mahwah, NJ: Lawrence Erlbaum Associates, Publishers (63-79).
- Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and biases. *Science*, 185, 1124-1131.
- Ziniel, S. (2008). "Cognitive Aging and Survey Measurement." Doctoral dissertation, University of Michigan.

October 9: No Class Meeting: Midterm exam due

October 16: Context Effects in Attitude Measurement

- Knäuper, B. & Schwarz, N. (2007). The Perils of Interpreting Age Differences in Attitude Reports: Question Order Effects Decrease with Age. *Journal of Official Statistics*, 23, 515–528.
- Schwarz, N. (2003). Culture-sensitive context effects: A challenge for cross-cultural surveys. In J. Harkness , F. van de Vijver, & P. Ph. Mohler (Eds.), *Cross-cultural survey methods* (pp. 93-100). New York: Wiley.
- Sudman, Bradburn & Schwarz (1996). Chapter 4, "Psychological sources of context effects in survey measurement," and Chapter 5, "The direction of context effects: What determines assimilation or contrast in attitude measurement?"

Recommended Reading

- Knäuper, B. (1998). Age differences in question and response order effects. In Schwarz, N., Park, D., Knäuper, B. and Sudman, S. (Eds.), *Cognition, Aging and Self-Reports*. Philadelphia: Psychology Press, pp. 341-363.
- McCabe, A., & Brannon, L. (2004). Application of conversational norms to the interpretation of survey results as a function of participants' need for cognition. *The Journal of Psychology*, 138, 91-94.
- Schuman, H. (1992). Context effects: State of the art/state of the past. In N. Schwarz & S. Sudman (Eds.), *Context effects in social and psychological research*. New York: Springer- Verlag, pp. 5-20.

October 23: Mapping and Reporting Answers

- Krosnick, J., & Alwin, D. (1987). An evaluation of a cognitive theory of response order effects in survey measurement. *Public Opinion Quarterly*, 51, 201-219.
- Schaeffer, N. C. (2000). Asking questions about sensitive topics: A selective overview. In Stone, A., Turkkan, J., Bachrach, C., Jobe, J., Kurtzman, H. and Cain, V. (Eds.), *The science of self-report*. Mahwah, NJ: Lawrence Erlbaum Associates, Publishers (105-121).
- SB&S (1996). Chapter 6, "Order effects within a question: Presenting categorical response alternatives."

Recommended Reading

- Johnson T.P., Shavitt S., Holbrook A.L. Survey response styles across cultures (2011). In Matsumoto D and van de Vijver FJR eds, (130-178). Cross-cultural research methods in psychology. Cambridge, UK: Cambridge University Press; 2011
- Knäuper, B. (1999). The impact of age and education on response order effects in attitude measurement. *Public Opinion Quarterly*, 63, 347-370.
- Schwarz, N. (1996). Chapter 5, "The conversational relevance of formal features of questionnaires" in *Cognition and communication: Judgmental biases, research methods, and the logic of conversation*. Mahwah, NJ: Lawrence Erlbaum Associates, Publishers.
- Schwarz, N., Hippler, H. J., Deutsch, B., & Strack, F. (1985). Response categories: Effects on behavioral reports and comparative judgments. *Public Opinion Quarterly*, 49, 388- 395.
- Tourangeau, R., Couper, M. P. & Conrad, F.G. (2004). Spacing, position and order: Interpretive heuristics for visual features of survey questions. *Public Opinion Quarterly*, 68, 368-393.
- Tourangeau, R., Couper, M. P. & Conrad, F.G. (2007). Color, labels and interpretive heuristics for response scales. *Public Opinion Quarterly*, 71, 91-112.

October 30: Classification and Coding

- Conrad, F.G., Couper, M.P., & Saksaugh, J. W. (2016). Classifying open-ended reports: Coding occupation in the current population survey. *Journal of Official Statistics*, 32, 75-92.
- Malhotra, N. & Krosnick, J. (2007). Procedures for updating classification systems: A study of biotechnology and the standard occupational classification. *Journal of Official Statistics*, 23, 409-432.

Recommended Reading

- Barsalou, L. (1991). Chapter 2, "Categorization." *Cognitive psychology: An overview for cognitive scientists*. Hillsdale, NJ: Lawrence Erlbaum Associates, Publisher.
- Cantor, D. & Esposito, J. (1992). Evaluating interviewer style for collecting industry and occupation information. *Proceedings of the Section on Survey Research Methods* (661-666).
- Gibson, J.L. & Caldeira, G.A. (2009). Knowing the Supreme Court? A Reconsideration of Public Ignorance of the High Court *The Journal of Politics*, 71, 429–441.
- Hak, T. and Bernts, T. (1996). Coder training: Theoretical training or practical socialization? *Qualitative Sociology*, 19, 479 -501.
- Kunda & Oleson (1995). Maintaining stereotypes in the face of disconfirmation: Constructing grounds for subtyping deviants. *Journal of Personality and Social Psychology*, 8, 565-579.
- Martin, J., Bushnell, D., Campanelli, P. and Thomas, R. (1995). A comparison of interviewer and office coding of occupations. *American Statistical Association: Proceedings of the Section on Survey Research Methods* (1122-1127).
- Zarató, M. and Sandoval, P. (1995). The effects of contextual cues on making occupational and gender categorizations. *British Journal of Social Psychology*, 34, 353-362.

November 6: Technology Mediated Responding

- Kreuter, F., Presser, S., and Tourangeau, R. (2008). Social desirability bias in CATI, IVR, and Web surveys: The effects of mode and question sensitivity. *Public Opinion Quarterly*, 72, 847-865.
- Lind, L.H., Schober, M.F., Conrad, F.G., & Reichert, H. (2013). Why do survey respondents disclose more when computers ask the questions? *Public Opinion Quarterly*, 77 888-935.
- Tourangeau, R. & Smith, T.W. (1998) Collecting sensitive information with different modes of data collection. In Couper, M. P., Baker, R. P., Bethlehem, J., Clark, C., Martin, J., Nicholls, W. L., O'Reilly, J. (Eds.) *Computer Assisted Survey Information Collection*. New York: New York: John Wiley & Sons, 431-453.

Recommended Reading

- Conrad, F.G., Couper, M.P., Tourangeau, R. & Peytchev, A. (2006). Use and non-use of clarification features in web surveys. *Journal of Official Statistics*, 22, 245-269.
- Conrad, F.G., Schober, M. F., & Coiner, T. (2007) Bringing features of human dialogue to web surveys. *Applied Cognitive Psychology*, 21, 165-188.

- Couper, M. P. (2008). Technology and the Survey Interview/Questionnaire. In Conrad, F.G. & Schober, M.F. (Eds.) *Envisioning the Survey Interview of the Future* (pp. 58-76). New York: Wiley & Sons.
- Couper, MP., Tourangeau. R., Marvin, T. (2009). Taking the audio out of Audio-CASI. *Public Opinion Quarterly*, 73, 281-301.
- Greatbatch, D., Luff, P., Heath, C., Champion, P. (1994). Interpersonal communication and human-computer interaction: An examination of the use of computers in medical consultations. *Interacting with Computers*, 5: 193-216.
- Lawrence, D., Atwood, M., Dews, S. and Turner, T. (1995). Social interaction in the use and design of a workstation: Two contexts of interaction. In Thomas, P. (Ed.), *The Social And Interactional Dimensions Of Human-Computer Interfaces*. New York: Cambridge University Press, pp. 240-259.

November 13: Satisficing

- Conrad, F. G., Couper, M. P., Tourangeau, R., & Zhang, C. (2017). Reducing speeding in web surveys by providing immediate feedback. In *Survey Research Methods*, 11, 45-67.
- Krosnick, J. 1991. Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied Cognitive Psychology*, 5, 213-236.
- Schober, M.F., Conrad, F.G., Antoun, C., Ehlen, P., Fail, S., Hupp, A., Johnston, M., Vickers, L., Yan, H.Y., Zhang, C. (2015). Precision and disclosure in text and voice interviews on smartphones. PLOS ONE.

Recommended Reading

- Holbrook, A.L., Krosnick, J.A., Moore, D. and Tourangeau, R. (2007). Response Order Effects in Dichotomous Categorical Questions Presented Orally: The Impact of Question and Respondent Attributes. *Public Opinion Quarterly* 71, 325– 48.
- Kapelner, A., & Chandler, D. (2010). Preventing Satisficing in online surveys. In *Proceedings of CrowdConf 2010*
- Roberts, C., Gilbert, E., Allum, N., & Eisner, L. (2019). Research synthesis: Satisficing in surveys: A systematic review of the literature. *Public Opinion Quarterly*, 83(3), 598-626.
- Zhang, Chan, and Frederick G. Conrad. (2013). Investigation of speeding in web surveys: Tendency to speed and its association with straightlining. *Survey Research Methods*, 8, 127-135.

November 20: Cognition and Cross-Cultural Survey Research

Final exam available after class

Schwarz, N., Oyserman, D., & Peytcheva, E. (2010). Cognition, communication, and culture: Implications for the survey response process. *Survey methods in multinational, multiregional and multicultural contexts*, 177-190.

Recommended Reading

Scheuch, E. K. (1968). The cross-cultural use of sample surveys: Problems of comparability. In S. Rokkan (Ed.), *Comparative research across cultures and nations* (pp. 176-209). The Hague, The Netherlands: Mouton.

November 27: Thanksgiving Holiday -- No Class Meeting

December 4: Final Exam Due by 11:30 AM; No Class Meeting